Listing status, host distribution, and human notifications for 30 zoonotic viruses in Korea, 2020–2025

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Background / Aims

- **Viral zoonoses** remain an important and evolving concern for both animal and human health, highlighting the need for coordinated surveillance and disease prioritization.
- WOAH and WHO prioritize zoonotic diseases under different animal-health and public-health frameworks.
- However, global prioritization may not necessarily correspond to **country-specific epidemiological contexts**, including the distribution of reservoir hosts and observed disease burden.
- We examined 31 zoonotic viruses with their hosts, national surveillance coverage and human burden, built a domestic risk index, and compared it with published risk-ranking tools.

Methodology

- **Viruses:** on WOAH Terrestrial Code Ch.1.3 (2025), the WHO 2024 prioritization, or the Korean notice / recorded domestic occurrence. Listing checked against primary texts.
- **Variables:** livestock head counts (KOSIS 2026 Q2), wildlife density survey coverage (NIBR 2025), human notifications 2020–2025 (KDCA), Korean GenBank records.
- **Domestic index:** mean of three axes ranked 0–1: bridge-host herd size, reservoir survey coverage (target species / under-sampled / outside the survey), notified human cases.

Study viruses, animal hosts in Korea

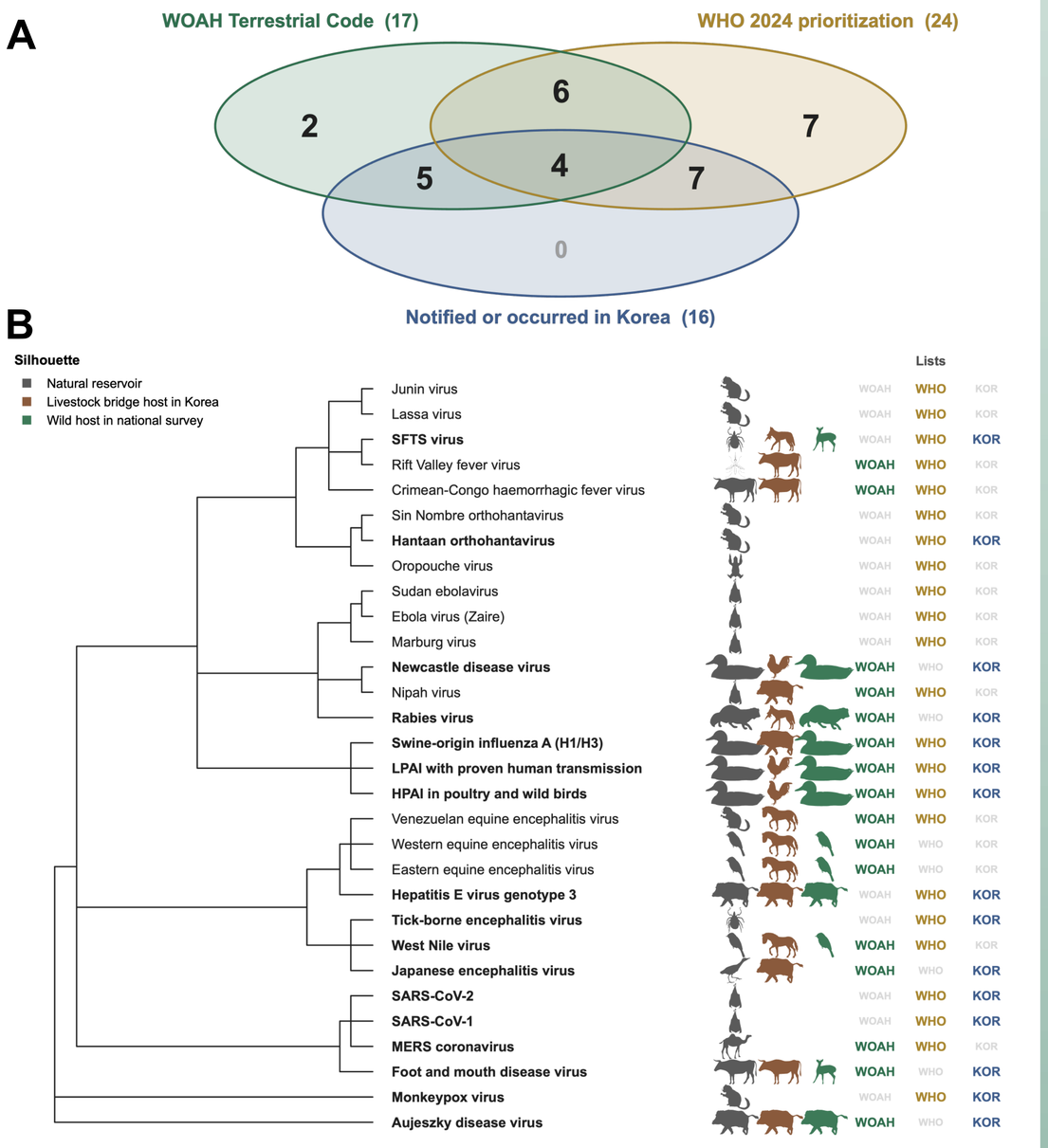


Fig 1. Study viruses and their associated host categories. (A) Overlap of viruses included in the WOAH and WHO lists and viruses reported in Korea. (B) Taxonomic distribution of the 30 viruses and their natural reservoir, livestock bridge, and surveyed wild hosts in Korea. Bold indicates viruses with an established reservoir or reported infection in Korea.

Result

Genome records by source

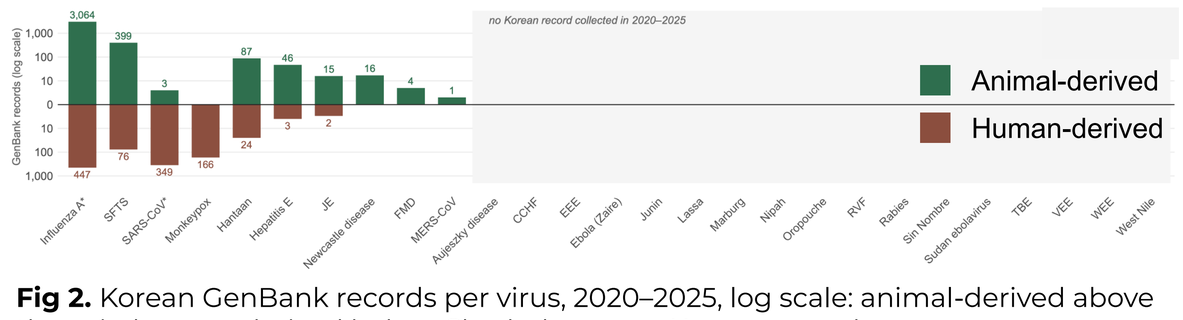


Fig 2. Korean GenBank records per virus, 2020–2025, log scale: animal-derived above the axis, human-derived below. Shaded area: no Korean record.

Risk Assessment

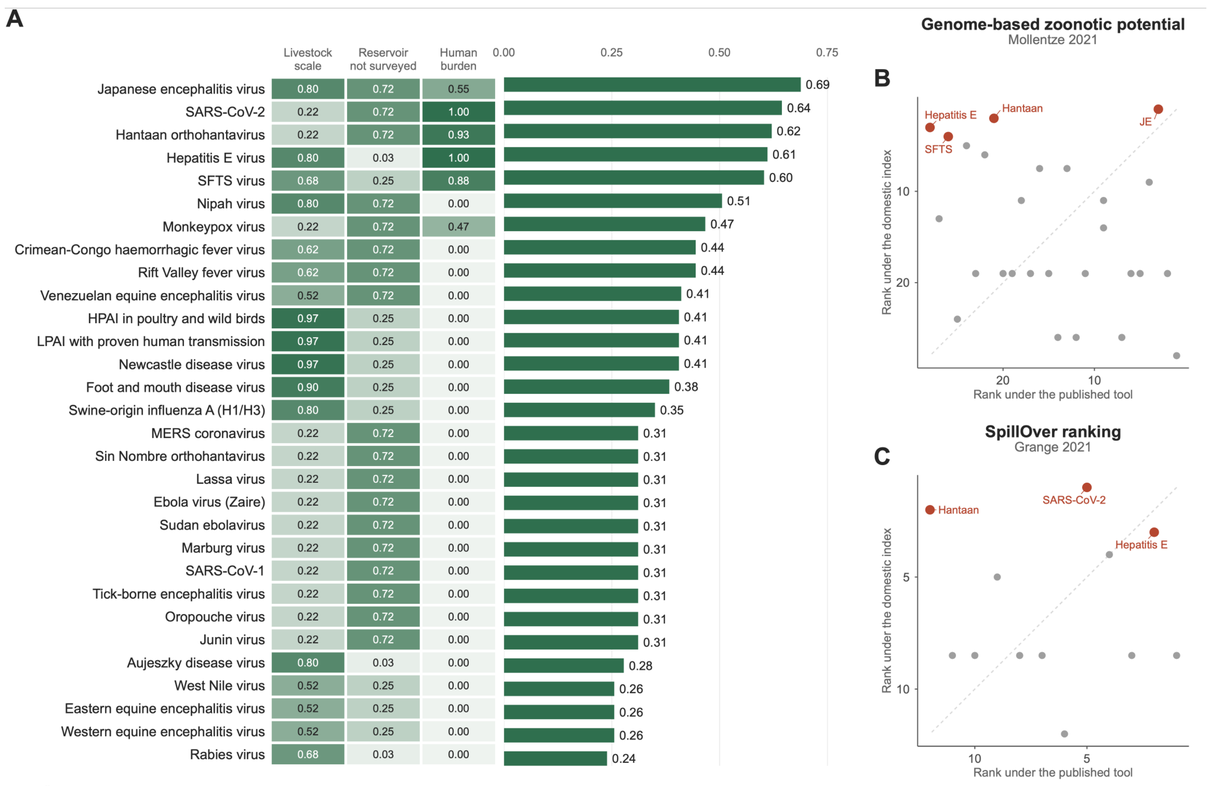


Fig 3. Domestic risk profiles of the 30 zoonotic viruses. (A) Livestock scale and reservoir surveillance gap are rank-normalised to 0–1, human burden as log10(cases + 1) over its maximum; bars are the mean of the three. (B, C) Domestic rank against genome-based zoonotic potential[3] and spillover risk[4]; red marks Korea's top five.

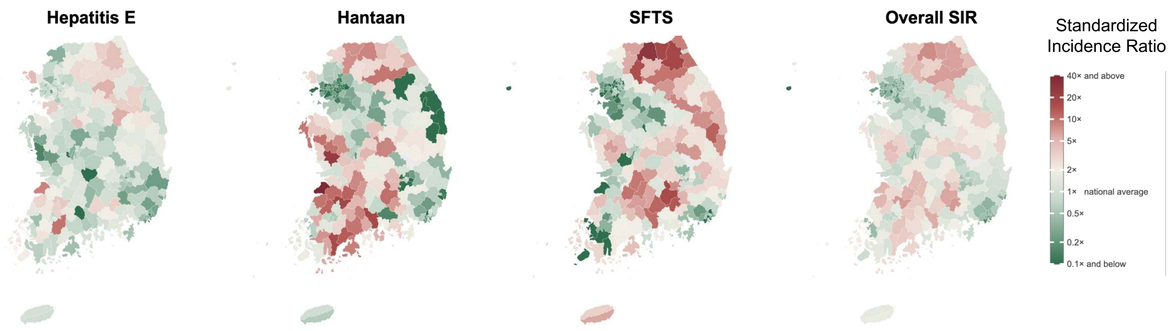


Fig 4. Standardized incidence ratio(SIR) by district, 2020–2025, for hepatitis E, Hantaan (HFRS), SFTS, and the combined SIR(geometric mean). 1× is the national rate.

Discussion

- **Sixteen** of the 30 viruses prioritized by WOAH and WHO **have been reported in Korea**; 17 have domestic livestock hosts, and wild hosts of 12 are covered by current wildlife surveillance.
- Integrating livestock population, reservoir surveillance and human cases ranked **risk of** JEV, SARS-CoV-2, Hantaan virus, HEV and SFTS virus highest; on recent human burden and infection ecology, **Hantaan, HEV and SFTS stand out domestically.**
- The **domestic risk ranking differed substantially from published global risk metrics**, which estimate general spillover potential rather than country-specific host distribution, surveillance coverage and observed burden.
- Hantaan virus, HEV and SFTS are spatially heterogeneous, with **high SIRs across Gangwon and Jeonnam**, the regions to target for surveillance first.

Conclusion

- Ranking livestock scale, surveillance coverage and human burden puts **Hantaan virus, HEV and SFTS** first among the 30 listed viruses.
- Global risk metrics rank them differently because they carry **no national context**.
- Surveillance must be **extended to rodents**, starting in Gangwon and Jeonnam, and the density survey's target-species list revised.

References

1) WOAH. Terrestrial Animal Health Code, Ch. 1.3. 2025.
2) WHO. Pathogens prioritization: a scientific framework for epidemic and pandemic R&D preparedness. 2024.
3) Mollentze N, et al. PLoS Biol. 2021;19:e3001390.
4) Grange ZL, et al. PNAS. 2021;118:e2002324118.

Acknowledgement

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